

# Mahesh Chitakoti

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## Summary

**Backend & AI Engineer** with **2.5+ years of experience** building and scaling high-throughput microservices, RAG systems, and event-driven GenAI pipelines. Specialized in **Python, FastAPI, TypeScript, Qdrant, and distributed caching**, with a proven track record of optimizing LLM token economics, reducing database latency, and delivering resilient, production-grade cloud APIs.

## Technical Skills

**Languages & Backend:** Python, TypeScript, SQL, FastAPI, Node.js, NestJS, Express.js, AsyncIO, REST APIs, Microservices, WebSockets, MQTT

**GenAI & Orchestration:** RAG, Agentic Workflows (LangChain, LangGraph), Prompt Engineering, Structured Outputs (Pydantic), Semantic Caching, Prompt Evaluation

**Vector Systems & Retrieval:** Qdrant, pgvector, Hybrid Search (Dense + BM25), Cross-Encoder Reranking, Embeddings Pipeline Optimization

**Cloud, DevOps & Data:** PostgreSQL, Redis, Docker, AWS (EC2, S3), GitHub Actions CI/CD, Google Gemini APIs, OpenAI API

## Experience

### AI Developer – SahioNext Technology Pvt. Ltd.

Oct 2025 – Apr 2026

*Remote*

- Architected an asynchronous GenAI pipeline using **FastAPI, Google Gemini APIs, and Qdrant**; implemented **semantic caching, Pydantic schema validation, and token-bucket rate limiting**, cutting redundant inference costs by **35%** and preventing rate-limit (429) throttling.
- Developed semantic-search microservices over lesson transcripts in **Qdrant**, implementing **hybrid retrieval (dense + BM25)** across **10,000+ chunks** to achieve **~300ms p95 latency** for automated CEFR language proficiency grading.
- Engineered full-duplex **WebSocket and MQTT** ingestion pipelines with **exponential backoff, dead-letter queues (DLQ), and automatic session recovery**, maintaining resilient real-time audio streaming across high-latency client connections
- Instrumented backend telemetry pipelines to monitor real-time token spend, rate-limit quotas, and pipeline bottlenecks; exposed operational metrics to monitoring dashboards, **reducing issue triage and debugging time from hours to minutes**.

### Full Stack Developer (Contract) – PearlThoughts

Feb 2025 – Sep 2025

*Remote*

- Engineered multi-tenant backend automation microservices using **NestJS, TypeScript, and Google Workspace APIs**, orchestrating **1,000+ daily business workflows** and eliminating manual client data-entry bottlenecks.
- Built distributed workflow pipelines with **exponential backoff, idempotent webhook handlers, and rate limiting**, ensuring resilient, idempotent job execution and preventing duplicate transaction processing across tenants.
- Developed automated verification harnesses and isolated **Docker autograder environments** for enterprise **Coursera course launches**, preventing runtime execution regressions prior to production rollout.

### Full Stack Developer – The DevOps Team

Oct 2023 – Jan 2025

*Bangalore, India*

- Optimized high-traffic **Node.js and Express.js APIs** using **Redis response caching, PostgreSQL B-tree indexing, and connection pooling**, reducing median response time from **800ms to 320ms (60% drop)**.
- Developed candidate-job recommendation workflows utilizing multi-attribute compound indexing in **PostgreSQL**, cutting search filter latency and improving relevant role matching accuracy by **22%**.
- Built **React.js and Next.js** onboarding experiences with integrated **A/B testing**, lifting user signup conversion rates by **18%**.

## Projects

### WorkBridge – AI-Powered Collaboration Platform

May 2026 – Present

*React.js, Next.js, Node.js, Express.js, Python, FastAPI, PostgreSQL, pgvector, Redis, OAuth | Live Project*

- Designed and implemented a decoupled microservices platform featuring a **Node.js/Express API gateway** for session/auth handling and a **Python/FastAPI microservice** for vector processing.

- Built core REST APIs with **PostgreSQL and Redis**, enforcing role-based access control (RBAC) and OAuth2 authentication to achieve **150ms p95 latency in k6 benchmark load tests**.
- Implemented candidate-to-project recommendation using **Python and pgvector cosine similarity**, delivering sub-second ranking retrieval over candidate skill embeddings.

Certifications

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| <b>Oracle Agentic AI Certified Foundations Associate</b> – Oracle                     | Aug 2026 |
| <b>Oracle AI Vector Search Certified Professional</b> – Oracle                        | Sep 2025 |
| <b>Machine Learning I</b> – Columbia University                                       | Aug 2025 |
| <b>Oracle Cloud Infrastructure 2024 Generative AI Certified Professional</b> – Oracle | Aug 2024 |

Education

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|----------------------------------------------------------|---------------------------|
| <b>Adichunchanagiri Institute of Technology (VTU)</b>    | Chikkamagaluru, KA, India |
| <i>Bachelor of Engineering in Mechanical Engineering</i> | Aug 2023                  |